2 June 1959
SUPERSEDING
NAVORD OS 5902A
24 May 1955

MILITARY SPECIFICATION

TEST SET MARK 27 MOD 1

1. SCOPE

1.1 This specification establishes the requirements for the procurement of Test Set Mark 27 Mod 1. The test set tests Batteries BA-239/U and BA-205/U and the overall operation of underwater Mines Mk 25 Mod 0, Mk 39 Mod 0 & Mk 49 Mod 0.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	Boxes, Fiber.

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-B-131	Barrier Material; Water Vapor-proof, Flexible.
MIL-V-173	Varnish, Moisture-and Fungus Resistant for the Treatment of Communications, Electronic and Associated Electrical Equipment.
MIL-D-3464	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification.

FSC 6625

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MIL-S-6872	Soldering Processes, General Specifications for.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWINGS

BUREAU OF ORDNANCE

LD 274969	Test Set Mark 27 Mod 1.
LD 284866	Preparation for Delivery for Test Set Cable Assemblies.
1476194	Marking Instructions for Storage and Shipping Containers.

And all documents listed in the above LD's and drawing.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400	General Instructions for the Design, Manufacture and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 General requirements.

3.1.1 Test Set Mark 27 Mod 1 shall be manufactured in accordance with LD 274969 and all documents listed thereon.

3.1.2 The requirements and procedures established in OP 400 shall be complied with as applicable.

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3.1.3 Preproduction sample. - Unless otherwise specified in the contract or order, a preproduction sample shall be manufactured by the same procedures and processes and at the same location as proposed for full production. This sample will be tested for the purpose of determining compliance with the requirements of this specification, the drawings, and the contract. See 4.1.2.

3.1.4 Interchangeability. - Component parts of assembled units (unless otherwise specified on the applicable drawings) fabricated in strict agreement with their applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the fabricated parts of a unit, selection or fitting is required and changes in dimensions are necessary, no changes in the parts as delineated on applicable drawings shall be made without prior approval of the Bureau of Ordnance.

3.1.5 Conflicting requirements. - Conflicting requirements arising between this specification and any other specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.1.6 Soldering. - Unless otherwise specified, soldering processes shall conform to the requirements specified in Military Specification MIL-S-6872. Only non-corrosive and non-volatile fluxes shall be used. Rosin shall be used when soldering wires to contacts and terminals. The use of rosin core solder is permitted. Electrical connections shall be mechanically secure and electrically continuous before soldering. Sufficient heat shall be applied during the soldering process to avoid "cold" solder joints.

3.1.7 Wiring. - All internal wiring shall be cabled wherever possible.

3.1.8 Moisture - and fungus-resistant treatment. - Materials and parts of this test set which are not moisture and fungus resistant shall be treated, before assembly, with a varnish conforming to Military Specification MIL-V-173.

3.2 Performance requirements and product characteristics. - The test set covered by this specification shall meet the following performance requirements and product characteristics.

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3.2.1 Meter, 300 volt range. - The test set meter shall read accurately within ± 6 volts, any voltage up to 300 volts impressed across test set terminals 300V and -, when tested as specified in 4.6.3.

3.2.2 Meter, 3 volt range. - The test set meter shall read accurately within ±0.06 volts, any voltage up to 3 volts impressed across the test set terminals 3V and -, when tested as specified in 4.6.4.

3.2.3 Meter, 30 ampere range. - The meter shall read accurately within ±0.6 amperes any current up to 30 amperes impressed across the test set terminals 30 AMPS and -, when tested as specified in 4.6.5.

3.2.4 Exciter circuit. - The exciter circuit shall be capable of delivering a variable current up to 120 milliamperes at the test set EXCITER terminals. The current delivered shall be measured on the milliampere scale of the test set meter (M-1) to an accuracy of ±3 milliamperes, when tested as specified in 4.6.6.

3.2.5 Dummy load. - With the test set adjusted to pass 120 milliamperes through the exciter coil, the current value shall not change over ±2 milliamperes when this current is switched to the dummy load. Test as specified in 4.6.7.

3.2.6 Load resistor. - The resistance of the load resistor (R-1) shall be 10,000 ±100 ohms as measured at test set terminals +300V and -, when tested as specified in 4.6.8.

3.2.7 Firing indicator circuit. - The firing indicator circuitry shall be properly connected as verified by testing as specified in 4.6.9.

3.3 Environmental requirement.

3.3.1 Vibration, rigidly mounted. - Test sets covered by this specification shall be mounted and vibrated as specified in 4.7.1, and thereafter shall show no damage and shall meet all performance requirements and product characteristics of this specification.

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3.4 Workmanship and cleanliness. - The test sets shall be free from dust, dirt, metal filings, moisture, solder splashes, chips and other foreign matter. Necessary precautions shall be taken during manufacture and assembly to insure these conditions.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection will be in accordance with the provisions of MIL-STD-105.

4.1.1 Lot. - As applied to the Government inspection of units of product, the term "lot" shall mean "inspection lot", i.e. a collection of units of the product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in "inspection lot" shall be as determined by the Government inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment or other purpose.

4.1.2 Preproduction sample. - Unless otherwise specified for each contract or order, a representative sample of one test set shall be manufactured using the methods and procedures proposed for full production. The Government inspector shall perform, witness or participate in such inspections and tests on the samples as are required for full production. Upon conditional acceptance by the inspector, the preproduction sample shall be forwarded, together with a report of inspection, to an activity designated in the contract or by the procuring agency, for the purpose of determining compliance with the requirements of the contract, specifications and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of the tests on preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.2.1 Preproduction sample for a subsequent contract. The necessity for a preproduction sample will be determined by the procuring agency when production under a new contract by the same contractor at the same location follows the manufacture of satisfactory material as covered by this document.

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4.1.3 Periodic production samples.

4.1.3.1 Unless otherwise specified, a random sample of test sets will be selected in accordance with the following table by the Government inspector and forwarded to an activity designated by the Bureau of Ordnance.

TABLE I

Test Sets Produced	Size of Sample
First 10	1
Next 10	1
Next 50	1
Every 50 thereafter	1

4.2 Inspection.

4.2.1 Government inspection. - The Government inspector will make such inspections as are necessary to determine that all components and assemblies are in accordance with the requirements of the contract, pertinent specifications and drawings. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in OP 400.

4.2.2 Contractor's inspection. - Contractor's inspection shall be conducted in accordance with provisions of "Inspection by Contractor" as contained in OP 400.

4.3 Classification of tests. - The inspection and testing of test set covered by this document shall be classified as follows:

(a) Preproduction tests
(b) Acceptance tests
(c) Periodic production tests

4.3.1 Preproduction and periodic production tests. - Preproduction and periodic production tests are those performed on samples submitted in accordance with 4.1.2 and 4.1.3 to determine that the lot and production meet all the requirements of this specification. Acceptable preproduction and periodic production sample units shall not be released for service use and shall be stamped or marked as follows: "Sample-Not for Service Use".

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4.3.2 Acceptance tests. - Acceptance tests for acceptance of the test set shall consist of tests to determine compliance with all requirements specified in this document with the exception of the environmental criteria.

4.4 Criteria for acceptance.

4.4.1 Preproduction tests. - Acceptance of preproduction sample shall be based on no defects in the sample.

4.4.2 Acceptance tests. - Failure of any test set to pass the acceptance tests shall result in rejection of the test set. If the number of failing test sets exceeds the number permitted by the associated criteria, the lot represented will be rejected.

4.4.3 Periodic production tests. - Acceptance of periodic production samples shall be based on no defects in the sample. Failure of a periodic production sample to comply with the requirements of this specification will result in rejection of all test sets submitted subsequent to the last acceptable sample until remedial measures have been taken to correct the conditions causing failure. When the faults have been corrected, sampling for periodic production testing shall revert to the top of Table I under 4.1.3.1. The manufacturer may resubmit rejected lots, subsequent to the completion of satisfactory corrective action as determined by the procuring activity.

4.5 Test equipment.

4.5.1 Test and inspection equipment and facilities. - Unless otherwise specified, the manufacturer shall furnish and maintain all necessary test equipment, facilities and personnel for performing all acceptance tests under the supervision of the Government inspector. The test equipment shall be adequate in quantity, and unless definite requirements are specified, they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.5.1.1 Equipment required for performing these tests is as follows:

(a) d-c voltmeter, 0-300 volt range, accuracy within ± 1/2 percent.

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(b) d-c voltmeter, 0-3 volt range, accuracy within ± 1/2 percent.

(c) d-c ammeter, 0-30 ampere range, accuracy within ± 1/2 percent.

(d) d-c milliammeter, 0-150 ma range, accuracy within ± 1/2 percent.

(e) Ohmmeter capable of measuring 10,000 ohms (10K) within an accuracy of ± 1 percent.

(f) Resistor, 2.6 ohms, accuracy within ± 5 percent.

(g) 50-300 volt variable d-c voltage source.

(h) 0-3 volt variable d-c voltage source.

(i) 0-30 ampere variable d-c current source.

4.6 Acceptance tests.

4.6.1 Test conditions. - Unless otherwise specified, tests shall be performed at 80° ±10°F and ambient relative humidity.

4.6.2 Visual inspection. - The test set and its component parts shall be inspected to verify that the materials, design, construction, physical dimensions, soldering, and workmanship are in accordance with this document and the drawings.

4.6.3 Meter, 300 volt range test. - To determine compliance with 3.2.1, connect the 0-300 volt d-c voltmeter (test item 4.5.1.1 (a)) together with the 50 to 300 volt d-c source (test item 4.5.1.1 (g)) across the test set terminals +300v and -. Set the CURRENT-OPER-VOLTS switch (SW-1) on the VOLTS position and the 1-2 switch (SW-3) on 1. Adjust the variable voltage source so that the test set meter indicates 100, 200, and 300 volts. When compared with the readings of the external meter, the test set meter shall be accurate to within ±6 volts.

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4.6.4 Meter, 3 volt range test. - To determine compliance with 3.2.2, connect the 0-3 volt d-c voltmeter (test item 4.5.1.1 (b)) together with the 0-3 volt d-c source (test item 4.5.1.1 (h)) across the test set terminals +3 volts and -. Set the CURRENT-OPER-VOLTS switch (SW-1) on the VOLTS position. Adjust the variable voltage source so that the test set meter indicates 1, 2, and 3 volts. When compared with the readings of the external meter, the test set meter shall be accurate to within ±0.06 volts.

4.6.5 Meter, 30 ampere range test. - To determine compliance with 3.2.3, set the CURRENT-OPER-VOLTS switch (SW-1) at CURRENT. Connect the 0-30 ampere d-c source (test item 4.5.1.1 (i)) and the 0-30 ampere d-c ammeter (test item 4.5.1.1 (c)) in series between the test set terminals +30 AMPS and -. Adjust the variable current source so that the test set meter indicates 10, 20, and 30 amperes. When compared with the readings of the standard meter, the test set meter shall be accurate to within ±0.6 amperes.

4.6.6 Exciter circuit test. - To determine compliance with 3.2.4, set the CURRENT-OPER-VOLTS switch (SW-1) on OPER. Set the 1-2 switch (SW-2) on 1. Set the 1-DUMMY-2 switch (SW-3) on 1. Connect the 0-150 milliammeter (test item 4.5.1.1 (d)) across the test set EXC. COIL terminals. Depress SW-4 and adjust CURRENT ADJ. knob so that the test set meter indicates 30, 60, 90, and 120 milliamperes. (Note: If external meter indicates negatively, reverse the leads). When compared with the readings of the external meter, the test set meter shall be accurate to within ±3 milliamperes. Adjust CURRENT ADJ. knob so that the test set meter indicates zero, and release SW-4. Set the 1-DUMMY-2 switch (SW-3) on 2. Reverse the external meter leads on EXC. COIL terminals. Depress SW-4 and adjust CURRENT ADJ. knob so that the test set meter indicates 150 milliamperes. The standard meter shall also indicate 150 ±3 milliamperes.

4.6.7 Dummy load test. - To determine compliance with 3.2.5, set the CURRENT-OPER-VOLTS switch (SW-1) on OPER. Set the 1-2 switch (SW-2) on 1. Set the 1-DUMMY-2 switch (SW-3) on either 1 or 2. Connect the exciter coil to the test set. Depress switch SW-4 and adjust the CURRENT ADJ. knob (R-5) so that the test set meter (M-1) indicates 120 milliamperes. With the switch SW-4 depressed, observe the test set meter when the 1-DUMMY-2 switch is in each of the 3 positions. The indication in each case shall be 120 ±2 milliamperes.

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4.6.8 Load resistor test. - To determine compliance with 3.2.6, set the 1-2 switch (SW-2) at position 2 and the CURRENT-OPER-VOLTS switch at OPER. The resistance between test set terminals +300v and - shall be 10,000 ±100 ohms when measured with test item 4.5.1.1 (e).

4.6.9 Firing indicator circuit test. - To determine compliance with 3.2.7, the following test procedures are specified.

4.6.9.1 With the 1-2 switch (SW-2) set on 2 and the CURRENT-OPER-VOLTS switch (SW-1) set on OPER, the indicating lamp (M2) shall light and the meter needle shall give a negative kick and then a positive indication when the LAMP-DET switch (SW-5) is placed in the LAMP position.

4.6.9.2 With the 1-2 switch (SW-2) set on 2, the CURRENT-OPER-VOLTS switch (SW-1) set on OPER, and a voltage source of 3 volts (test item 4.5.1.1 (h)) in series with the 2.6 ohm resistor (test item 4.5.1.1 (f)), connected to the cable CA-288, the indicating lamp (M2) shall light and the meter needle shall give a negative kick and then a positive indication when the LAMP-DET (SW-5) switch is placed in the DET position. The cable terminal C shall be connected to the positive terminal of the voltage source.

4.7 Environmental test. - Test sets subjected to the environmental condition given below shall be tested thereafter to determine compliance with all other requirements of this specification.

4.7.1 Vibration, rigidly-mounted, test. - Mount the test set rigidly to a vibration table. The vertical displacement component of the vibration shall be simple harmonic motion. The sample shall be mounted on the vibration table so that equal quantities of test sets experience vibration perpendicular to the plane of mounting and vibration parallel to the plane of mounting. The units shall be vibrated through a range of frequencies from 700 to 3000 cycles per minute in 100 cycles per minute increments. The duration of the vibration of each increment shall be one hour. The amplitude of the vibration shall be in accordance with the following schedule.

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FREQUENCY (Cycles per minute)	AMPLITUDE (Inches)
700 to 1500	0.030 ± 0.002
1600 to 2000	0.020 ± 0.002
2100 to 2500	0.013 ± 0.001
2600 to 3000	0.009 ± 0.001

5. PREPARATION FOR DELIVERY

NOTE: This equipment is very fragile, contractor shall contact Bureau of Ordnance (MABlg) for special handling instructions).

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning and drying. - (Not applicable).

5.1.1.2 Preservative application. - (Not applicable).

5.1.1.3 Unit packaging. - Prior to unit packaging, test set cables shall be packaged in accordance with Bureau of Ordnance List of Drawings LD 284866 and positioned in cable well of test set. Each test set shall be initially packaged in a Style RSC fiberboard box fabricated from Classl, Type I or Type II board, having 275 lb. minimum bursting strength, conforming to the requirements of Federal Specification PPP-B-636. Sufficient fillers and folded corner pads, of the same box material, shall be utilized within the carton to provide a tight pack and to prevent direct contact between protrusions of the test set case and the walls of the fiberboard box.

5.1.1.3.1 Each test set, prepared as specified above, shall be further packaged Method II-b conforming to the requirements of Military Specification MIL-P-116. The inner carton, containing the packaged test set, shall be inserted into a water-vaporproof bag, fabricated from Class 1 barrier-material conforming to requirements of Military Specification MIL-B-131, and closure heat-sealed.

5.1.1.3.2 A Style RSC fiberboard box, Class 2, W5c or W5s compliance symbol, conforming to the requirements of Federal Specification PPP-B-636, shall be employed for the outer carton and designed to snugly enclose the barrier-sealed inner container. The required desiccant shall be suitable positioned within the test set case. For conformance and quantity required see Military Specification MIL-D-3464.

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5.1.2 Level B.

5.1.2.1 Cleaning and drying. - (Not applicable).

5.1.2.2 Preservative application. - (Not applicable).

5.1.2.3 Unit packaging. - Test set shall be prepared and packaged as specified under 5.1.1.3.

5.1.2.3.1 Each test set, prepared as specified above, shall be further packaged by Method IA-14, conforming to the requirements of Military Specification MIL-P-116. The inner carton, containing the packaged test set, shall be inserted into a water-vaporproof bag, fabricated from Class I barrier material conforming to requirements of Military Specification MIL-B-131, and closure heat-sealed.

5.1.2.3.2 A Style RSC fiberboard box, Class 2, W5c or W5s compliance symbol, conforming to the requirements of Federal Specification PPP-B-636, shall be employed for the outer carton and designed to snugly enclose the barrier sealed inner container. The required desiccant shall be suitable positioned within the test set case.

5.1.3 Level C.

5.1.3.1 Test set units shall be packaged in a manner which will afford adequate protection from deterioration and physical damage during shipment from the contractor or supply source to the first receiving activity for immediate use.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior containers. - Each test set, individually packaged as specified under 5.1.1, shall be packed in an overseas type wood-cleated plywood box or a Class 2, Style 2, 2-1/2, 3 or 4 nailed wood box conforming to the requirements of Federal Specification PPP-B-601 or PPP-B-621, respectively.

5.2.2 Level B.

5.2.2.1 Exterior containers. - Each test set, individually packaged as specified in 5.1.1 shall be packed in a domestic wood cleated plywood box, or a Class 1, Style 2, 2-1/2, 3 or 4 nailed wood box conforming to the requirements of Federal Specifications PPP-B-601 or PPP-B-621, respectively.

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5.2.3 Level C.

5.2.3.1 Exterior containers. - Each test set shall be packed in a manner which will provide protection from damage during direct shipment from the contractor or supply source to the first receiving activity for immediate use. The type of containers used shall conform to applicable carrier rules and regulations.

5.3 Marking.

5.3.1 Normal markings. - In addition to any special marking required by the contract or order, all marking shall be in accordance with Military Standard MIL-STD-129.

5.3.1.1 For details on marking requirements, see Bureau of Ordnance Drawing Number 1476194.

5.3.1.2 Special markings. - The following special marking shall appear on the top, one (1) side and one (1) end of each shipping container:

DELICATE INSTRUMENT
HANDLE WITH CARE

6. NOTES.

6.1 Ordering data. - Procurement documents should specify the title, number and date of this specification, and exceptions to this specification, applicable drawings and other documents. Procurement documents should also specify the desired level of packaging and packing.

6.2 Criteria for use of proper level of preservation, packaging and packing shall be as follows:

For Level A. - This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level B. - This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for 6 months or less or in domestic locations for an indefinite period.

For Level C. - This level shall be used only when it is definitely known that the packaged item is to be shipped to doemestic installations for immediate use at the first receiving activity.

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